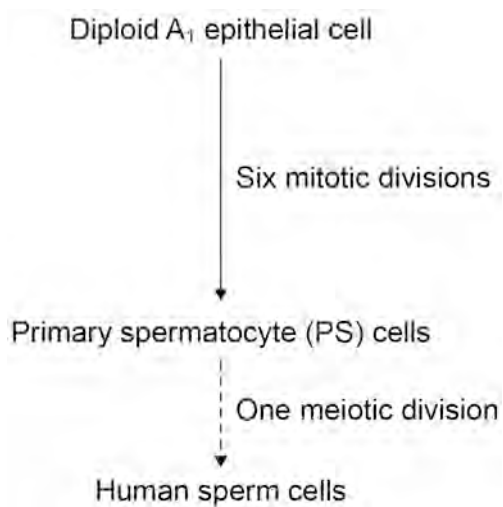


0 7 . 1

Figure 7 shows how human sperm cells are produced in the testes from diploid A₁ epithelial cells.

Figure 7



Use **Figure 7** to calculate the minimum number of A₁ cells needed to produce 1.5×10^7 sperm cells.

Give your answer in standard form.

Show your working.

[2 marks]

Answer _____ A₁ cells

0 7 . 2

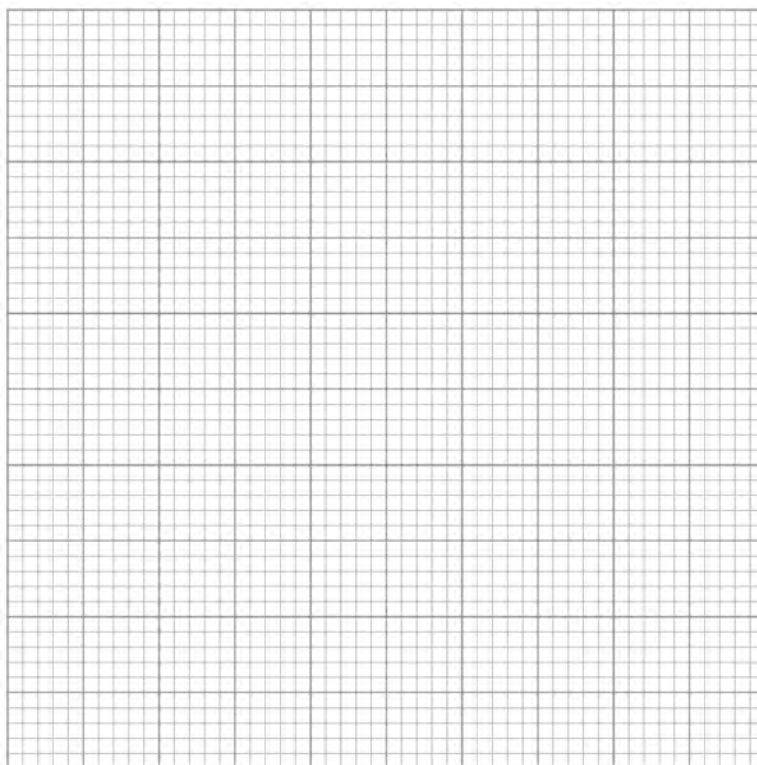
A PS cell goes through interphase before a meiotic division.

A scientist determined that the mass of DNA in **one** PS cell at the start of interphase is 8 arbitrary units.

Draw a bar chart in **Figure 8** showing the mass of DNA in **one** cell in each of these phases:

- at the start of interphase
- at the end of interphase
- at the end of the first meiotic division
- at the end of the second meiotic division.

Figure 8



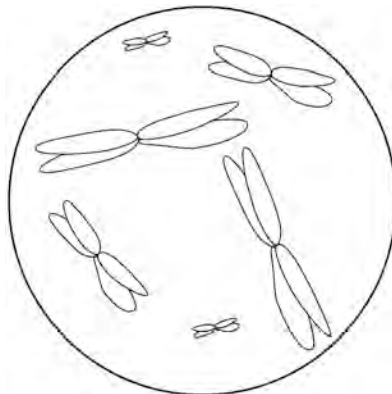
[3 marks]

Question 7 continues on the next page

0 7 . 3 A PS cell contains 46 chromosomes.

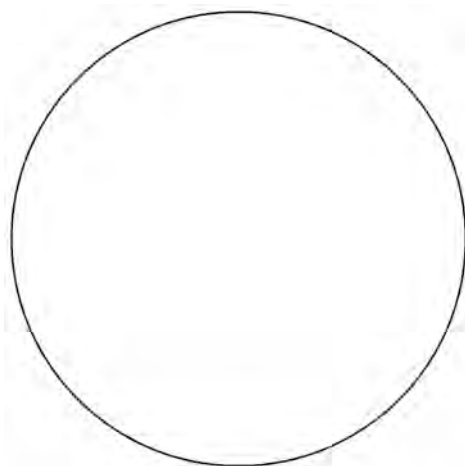
Figure 9 shows the appearance of six of those chromosomes just as a PS cell enters meiosis.

Figure 9



Using the information in **Figure 9**, draw chromosomes in **Figure 10** as they would appear in **one** cell at the end of the first meiotic division.

Figure 10



[1 mark]

0 7 . 4

A mutation can produce a gamete with **one** extra chromosome.

Name the type of mutation that produces a gamete with one extra chromosome.

Explain how meiosis can produce a gamete with one extra chromosome.

[3 marks]

Type of mutation _____

Explanation _____

Turn over for the next question

